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Smartphone 'addiction'? Evidence from withdrawal anxiety and salience-induced inhibitory control impairment

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ABSTRACT

This study aimed to explore whether problematic smartphone use (PSU) aligns with behavioural addiction by evaluating two key criteria: (1) higher withdrawal anxiety induced by smartphone separation and (2) impaired inhibitory control with neurobiological manifestations. A mixed-method approach compared PSU individuals (mean age = 21) and non-PSU (mean age = 20.33). Withdrawal anxiety was measured using the State-Trait Anxiety Inventory, and inhibitory control was assessed with a modified Go/NoGo task enhanced by Event-Related Potentials. Participants ($N = 30$) were categorised based on the Smartphone Addiction Scale-Short Version. The PSU individuals reported greater anxiety when separated from their smartphones, but the anxiety response was transient, peaking briefly before decreasing upon reunion. In terms of inhibitory control, PSU individuals showed significantly reduced NoGo-P3 amplitudes in response to smartphone-related cues, indicating impaired inhibitory control. However, no differences were observed in the NoGo-N2 phase. The PSU individuals exhibited signs of behavioural addiction, such as withdrawal anxiety and impaired inhibitory control, but these findings are insufficient to definitively categorise PSU as a behavioural addiction. Further research is needed to explore the neurobiological and psychological mechanisms underlying problematic smartphone use and its potential classification as an addiction.

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Problematic smartphone use; behaviour addiction; state anxiety; inhibitory control; event-related potentials

1. Introduction

The smartphone has integrated into nearly every aspect of human existence (Belk 2016; Clayton, Leshner, and Almond 2015). Notwithstanding the immense benefits, extensive research has documented negative consequences associated with excessive smartphone use, including anxiety, depression, sleep disturbances and cognitive decline (Chen et al. 2016; Li et al. 2020; Yang et al. 2020). These adverse effects have led to the concept of 'problematic smartphone use' (PSU), defined as an excessive reliance on smartphones that results in negative functional outcomes (Billieux et al. 2015).

PSU shares similarities with behavioural addiction (Billieux et al. 2015). Indeed, numerous studies have utilised the term 'smartphone addiction' to depict this phenomenon (Kwon et al. 2013a, 2013b; Wang, Zhou, and Hu 2024; Yu and Sussman 2020). The classification of problematic smartphone use as a 'behavioural addiction' remains a contentious issue among scholars (Yu and Sussman 2020). Proponents argue that the compulsive use of smartphones shares hallmark features with recognised

addictions, including withdrawal and tolerance (Bianchi and Phillips 2005; Billieux et al. 2015). Critics, however, caution against hastily categorising behavioural patterns without thorough evidence (Chóliz 2010; Elhai et al. 2017; James et al. 2023). This debate underscores the need for empirical studies to explore the neurobiological aspects of problematic smartphone use, specifically its alignment with the criteria established for behavioural addictions. As the purpose of this study is to investigate whether this phenomenon is a behavioural addiction, the neutral term 'problematic smartphone use' is employed.

1.1 Behavioural addiction and its criteria

Behavioural addiction (represented by gambling disorder in DSM-5) is characterised by a persistent pattern of repetitive behaviour that activates the brain's reward system (similar to drug abuse) and produces symptoms analogous to those seen in substance use disorders (American Psychiatric Association [APA] 2013). This behaviour continues despite individuals' attempts to stop or control

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their usage (Marlatt et al. 1988). The diagnostic criteria for addictive disorders are largely derived from those for substance disorders, encompassing mood modification, salience, tolerance, withdrawal, relapse and conflict (Griffiths 2005, 2017). These criteria are widely recognised and serve as a benchmark for identifying addictive behaviours (Cutino and Nees 2017; Fernandez, Kuss, and Griffiths 2020; Wilcockson, Osborne, and Ellis 2019).

The DSM-5's 2013 edition introduced 'Substance-Related and Addictive Disorders' as categories that include both substance-related disorders and non-substance-related (behavioural) addictions. This inclusion sparked a renewed interest in the research of behavioural addictions, yet, so far, only gambling disorder is officially recognised under this category. Many researchers urge caution in hastily labelling routine or recreational behaviours as pathological (Grant et al. 2014; Griffiths 2005, 2017; Kardefelt-Winther et al. 2017; Marlatt et al. 1988), highlighting the need for substantial empirical evidence before considering new behaviours as addictions.

Given this context, it is crucial to methodically evaluate the behaviours associated with smartphone use against the criteria for behavioural addiction. Due to the extensive criteria set for behavioural addiction, it is impractical to assess all aspects in a single study. Thus this study focuses on two critical criteria: withdrawal anxiety and salience-induced inhibitory control impairment.

This focus can be understood from the following theoretical perspectives. First, withdrawal anxiety captures the negative reinforcement mechanism central to addiction (Koob and Le Moal 2008) and is a DSM-5 diagnostic criterion for both substance use disorders and gambling disorder (APA 2013). Second, salience-induced inhibitory control impairment derives from the Impaired Response Inhibition and Salience Attribution (iRISA) model (Goldstein and Volkow 2002, 2011), which posits that addiction-related cues acquire heightened salience and disrupt inhibitory control. This cue-specific deficit is a hallmark of addiction, distinguishing it from domain-general inhibitory impairments (Luijten, Littel, and Franken 2011). This study examines two criteria: one affective (withdrawal anxiety) and one cognitive (salience-induced inhibitory control impairment). By doing so, it aims to provide empirical evidence regarding whether problematic smartphone use aligns with established neurobehavioural frameworks of addiction.

1.2 Withdrawal anxiety

Withdrawal symptoms encompass the physiological and psychological reactions that manifest when a specific activity is either ceased or significantly reduced (Bahji

et al. 2020; Fernandez, Kuss, and Griffiths 2020; Griffiths 2005). Among these symptoms, anxiety is a principal indicator as outlined in the DSM-5, signifying its critical role in understanding withdrawal phenomena (APA 2013; Brown 1993; Fauman 2002; Fernandez, Kuss, and Griffiths 2020; Hellem 2016; Sapacz, Rockman, and Clark 2016). This underscores the selection of withdrawal anxiety as a pivotal criterion for addiction analysis.

Recent years have seen an uptick in research exploring withdrawal anxiety resulting from restricted smartphone use (Anrijs et al. 2018; Brailovskaia et al. 2023; Cheever et al. 2014; Clayton, Leshner, and Almond 2015; Cutino and Nees 2017; Eide et al. 2018; Fernandez, Kuss, and Griffiths 2020; Hartanto and Yang 2016; Sapacz, Rockman, and Clark 2016; Wilcockson, Osborne, and Ellis 2019). However, most of these studies did not differentiate between participants based on their problematic smartphone use levels. Consequently, there is a scarcity of research directly comparing withdrawal anxiety in individuals identified with problematic smartphone use against those without such a diagnosis. This comparison is crucial for discerning whether problematic smartphone use aligns with the established frameworks of behavioural addiction.

While controlled studies in this area are limited, existing research presents mixed findings. Cheever et al. (2014) observed that only moderate-to-high users showed significant increases in withdrawal anxiety over time. By contrast, both Sapacz, Rockman, and Clark (2016) and Cutino and Nees (2017) found no evidence of such withdrawal-related anxiety increases. These discrepancies may stem from core differences in research participants, measurement protocols, experimental paradigms and research contexts across studies (see Table 1 for details). For instance, Cheever et al. (2014) required participants to be separated from their phones for 75 min in a windowless, clock-free and completely distraction-free lecture hall. Cutino and Nees (2017) asked participants to complete a 60-minute study task while their phones were taken away. By comparison, Sapacz, Rockman, and Clark (2016) only separated participants from their phones for approximately 3–5 min. To address these discrepancies, the present study continuously assessed withdrawal anxiety during phone separation, providing additional evidence to disentangle the effects of design heterogeneity.

1.3 Salience-induced inhibitory control impairment

Salience refers to the process by which a specific activity dominates an individual's thoughts, emotions, and

Table 1. Comparisons of three studies examining smartphone restriction and anxiety.

Dimension	Cheever et al. 2014	Cutino and Nees 2017	Sapacz, Rockman, and Clark 2016
Sample characteristics	Wide age range (19–57 years), gender-balanced; grouped by daily usage duration into low/moderate/high users	Narrow age ($\approx$ 19.5 years), female-majority; grouped by PMPU scale median into high/low PMPU groups	Narrow age (18–24 years), overwhelmingly female; randomly assigned to 4 groups by registration time
Experimental paradigm	Device taken away vs. turned off & out of sight; passive sitting with no task	Device surrendered vs. retained; 60-minute independent study	Four conditions (taken away, salient, hidden, free use); 3–5 min of waiting
Measures	STAI-State anxiety, measured 3 times at 20-minute intervals	STAI-State anxiety, pre- and post-test once each	STAI-State anxiety (single post-test); combined with social anxiety and addiction proneness scales
Contextual factors	Windowless, clock-free, distraction-free lecture hall; low ecological validity	Independent study setting; high ecological validity	Group waiting scenario; short duration, moderate ecological validity
Key findings	High/moderate PSU predicted significant increases in state anxiety under mobile phone restriction, while low PSU did not	High PSU was linked to higher baseline state anxiety, but phone restriction did not further increase their state anxiety	PSU severity was not associated with elevated state anxiety in response to mobile phone restriction

behaviours, becoming a central focus of their life (Griffiths 1996, 2005). The iRISA model provides insight into this phenomenon, explaining how cues associated with addiction gain prominence, leading to a decreased sensitivity toward natural, non-addictive rewards. This dynamic results in compromised regulatory control over addictive behaviours and diminished capacity for inhibiting negative decision-making processes (Goldstein and Volkow 2002, 2011). Evidence of this effect has been extensively documented in substance addiction, illustrating how such cues significantly disrupt cognitive control mechanisms (Heilig et al. 2021; Korucuoglu, Gladwin, and Wiers 2015; Liu et al. 2021; Luijten, Littel, and Franken 2011).

The mechanism of diminished decision-making control, as seen in substance addictions, is also applicable to behavioural addictions (Antons, Brand, and Potenza 2020; Brand et al. 2016, 2019). If problematic smartphone use is regarded as a type of addiction, it should exhibit salience-related features. To be more precise, individuals with problematic smartphone use tend to assign higher salience to smartphone-related cues, display reduced sensitivity towards neutral cues, and demonstrate a salience-induced inhibitory control impairment based on the iRISA-model.

Given the theoretical foundations established by the iRISA-model and the hypothesised parallels between substance and behavioural addictions, empirical investigation into the neurocognitive impairments associated with problematic smartphone use becomes paramount. This requires precise measurement techniques capable of capturing the nuanced dynamics of inhibitory control impairment induced by salience.

1.3.1. Measurement of inhibitory control

According to Diamond's (2013) seminal review of executive functions, inhibitory control can be subdivided into interference control and response inhibition. Interference control primarily involves selective attention and the

suppression of prepotent mental representations (cognitive inhibition), and is commonly measured by tasks such as the Flanker and Stroop tasks. Response inhibition, also referred to as self-control, centres on resisting temptations, inhibiting impulsive actions or overriding intuitive behavioural tendencies. This component of inhibitory control aligns closely with core deficits in addictive behaviours, such as the inability to resist cravings and impulsive responses to salient cues. The Go/No-go task is one of the most typical paradigms for measuring response inhibition. Although the Stop-signal task is also used to measure response inhibition, it assesses the ability to cancel a response that is already prepared or in progress, which reflects a form of reactive inhibition. This differs from the kind of proactive impulse control required in daily life, where individuals attempt to restrain urges before acting on salient, temptation-related stimuli. Therefore, the present study adopted a modified Go/No-go paradigm to better capture salience-induced impairments in response inhibition among individuals with problematic smartphone use.

This modified Go/No-go task serves as a commonly used tool in assessing inhibitory control impairments induced by salient stimuli, particularly within populations grappling with addiction (Chen et al. 2016; Detandt et al. 2017; Gao et al. 2020; Korucuoglu, Gladwin, and Wiers 2015; Luijten, Littel, and Franken 2011). This paradigm is designed to elicit a dominant response, requiring participants to quickly respond to 'Go' signals and restrain their response to the sporadic 'NoGo' signals. The challenge intensifies when attempting to inhibit the conditioned response to 'NoGo' signals, which is indicative of inhibitory control capabilities (Di Russo et al. 2006; Jonkman 2006).

In line with the iRISA model, this study posits that individuals with problematic smartphone use may show an increased sensitivity to smartphone-related cues, potentially impairing their ability to control impulsive reactions. To investigate this hypothesis, the

traditional Go/NoGo task has been adapted by incorporating stimuli associated with smartphones (or neutral images) as background cues for both 'Go' and 'NoGo' signals. This modification aims to discern whether the presence of smartphone-related cues, deemed highly salient to the problematic smartphone use individuals, compromises their inhibitory control.

1.3.2. ERP components of inhibitory control

To classify problematic smartphone use as a behavioural addiction, it is essential to integrate neurobiological measures (e.g. ERPs) with traditional behavioural assessments (Billieux et al. 2015; Brand et al. 2019). ERPs, detectable as time-locked electroencephalographic responses to specific stimuli or during information processing, offer a window into the neural underpinnings of cognitive functions (Luck 2005). Their sensitivity surpasses that of behavioural measures, potentially uncovering subtle cognitive differences not evident through behavioural observation alone (Chen et al. 2016; Gao et al. 2020; Korucuoglu, Gladwin, and Wiers 2015).

In the context of inhibitory control, the Go/No-go task has been instrumental in eliciting two critical ERP components: N2 and P3. These components are notably pronounced in the frontal-central regions of the brain, with the NoGo condition generating larger amplitudes than the Go condition, indicative of the neural effort involved in inhibitory processes (Chen et al. 2020; Donkers and van Boxtel 2004).

Specifically, the NoGo-N2 component, linked to the anterior cingulate cortex (ACC), is associated with the monitoring and resolution of conflicts from irrelevant stimuli. In the realm of addiction, the heightened salience attributed to addiction-related cues can amplify the NoGo-N2 amplitude, signalling increased conflict monitoring efforts (Chen et al. 2016; Falkenstein 2006; Korucuoglu, Gladwin, and Wiers 2015; Nieuwenhuis et al. 2003).

Furthermore, the NoGo-P3 component, related to the motor and premotor cortex, reflects the culmination of the motor inhibition process. Individuals with PSU always exhibit reduced NoGo-P3 amplitudes compared to controls, marking a neural deficit in motor inhibitory control. This reduction is especially pertinent in the presence of addiction cues, aligning with the iRISA model's predictions of diminished inhibitory control due to salience (Chen et al. 2020; Detandt et al. 2017; Gao et al. 2020; Luijten, Littel, and Franken 2011).

Collectively, the N2 and P3 components serve as distinct but complementary indicators of the cognitive processes underpinning inhibitory control. Their analysis not only provides evidence of the specific neural mechanisms affected by problematic smartphone use but also

reinforces the iRISA model's application to understanding behavioural addictions through the lens of neurobiological impairment.

Our review of the existing literature reveals that only a handful of studies, specifically two that are closely related to our research focus, have explored the impact of salience-induced inhibitory control impairment in individuals with problematic smartphone use employing the modified Go/NoGo task in conjunction with ERP measurements (Chen et al. 2016; Gao et al. 2020). These pioneering studies, while invaluable, present conflicting findings regarding the extent and nature of inhibitory control impairments in this population. Such discrepancies could be attributed to the four-year interval between the two studies, during which there was a rapid evolution in mobile technology and an increasing reliance on smartphones among the populace. It is imperative to conduct a follow-up study today, given the substantial advancements in smartphone-related technologies over the past years.

Despite these differences, both studies converge on the critical point that there is a discernible disruption in inhibitory control processes among individuals exhibiting signs of problematic smartphone use. This disruption, as indicated by alterations in ERP components, underscores the complexity of cognitive responses in the context of behavioural addiction. The inconsistent results, rather than detracting from the validity of these observations, instead highlight the urgent need for further empirical exploration. It is imperative to delve deeper into these cognitive processes, employing a variety of stimuli and methodological approaches to capture the full spectrum of inhibitory control impairments associated with smartphone use.

Our study aims to build upon these foundational findings, addressing the gaps and discrepancies observed. By employing a nuanced approach to the modified Go/NoGo paradigm and leveraging advanced ERP analysis, we seek to clarify the mechanisms underlying salience-induced inhibitory control impairments. This endeavour contributes to the academic discourse on problematic smartphone use and aids in developing targeted interventions by elucidating the neurocognitive dimensions of problematic smartphone use.

1.4 Present study

The primary objective of this study is twofold: first, to explore the anxiety experienced by individuals with problematic smartphone use during periods of withdrawal; and second, to evaluate the influence of smartphone cue salience on inhibitory control within this population. To measure withdrawal-induced anxiety, we adopted a

self-report methodology, utilising the State component of the State Anxiety Inventory (S-AI), a method inspired by previous research (Cheever et al. 2014; Cutino and Nees 2017; Sapacz, Rockman, and Clark 2016). Participants completed the S-AI during the experimental tasks to provide direct insights into their anxiety levels at various stages of smartphone access or restriction. For evaluating inhibitory control impairments influenced by the salience of smartphone cues, we implemented a modified Go/NoGo task combined with Event-Related Potential (ERP) technology. This approach is designed to measure both behavioural responses and underlying neurophysiological processes, drawing on methodologies previously applied to studies of substance addiction (Detandt et al. 2017; Korucuoglu, Gladwin, and Wiers 2015; Luijten, Littel, and Franken 2011).

Based on the above methodological foundation, we articulate the following hypotheses, directly aligned with the criteria for diagnosing behavioural addiction:

Hypothesis 1: Individuals with PSU will undergo higher anxiety compared to controls during smartphone withdrawal.

Hypothesis 1a: During the smartphone withdrawal, the PSU group will consistently exhibit higher levels of anxiety compared to the control group.

Hypothesis 1b: During the smartphone withdrawal, the PSU group would experience a sharp rise in anxiety upon separation from their phones, followed by a rapid decline upon reunion. In contrast, the control group would show less variation.

Hypothesis 2: Individuals with PSU will exhibit inhibitory control deficits in comparison to controls due to the salience of smartphone-related cues.

Hypothesis 2a: In comparison to the control group, individuals with PSU are expected to exhibit an increase in NoGo-N2 amplitude during smartphone background cue trials.

Hypothesis 2b: In comparison to the control group, individuals with PSU are expected to exhibit a decrease in NoGo-P3 amplitude during smartphone background cue trials.

2. Method

2.1 Participants

Four hundred twenty-one participants were recruited from the lessons at the local university and were asked to complete a survey focusing on mental health and daily smartphone usage.

Using G*Power 3.1, with a medium effect size of 0.25 and a power of 0.8, the minimum sample size was

calculated to be 24 for the analysis of behavioural and EEG data related to inhibitory control, and 28 for the analysis of withdrawal anxiety levels. Participants were divided into two groups based on the Smartphone Addiction Scale (SAS; Kwon et al. 2013b) and the SAS-Short Version (SAS-SV; Kwon et al. 2013a) scores. The PSU group ($n = 17$) included participants with SAS scores in the top 30% of the sample distribution (a threshold selected following the extreme-groups design; Fisher et al. 2020) who also met the SAS-SV addiction threshold. The NPSU group ($n = 17$, control) included participants with SAS scores in the bottom 30% who did not meet the SAS-SV addiction threshold. All 34 participants were willing to take part in the experiment. During the course of the experiment, four participants were excluded owing to the removal of more than 20% of their trials or due to excessive noise during EEG signal recording. Consequently, both the PSU and control groups consisted of 15 participants each (PSU group: 8 females, 7 males, mean age = 21 ± 1.77 years; control group: 7 females, 8 males, mean age = 20.33 ± 1.45 years). All participants were healthy individuals without any history of psychiatric or neurological disorders, possessing normal or corrected vision and being right-handed.

Upon completion of the experimental task, data regarding the participants' smartphone usage, including total daily usage and time spent on various smartphone activities, was collected. Smartphone use activities were classified into four types: interpersonal use, which refers to smartphone use for social communication and maintaining interpersonal relationships, such as using Facebook; entertainment use, which involves smartphone use for leisure and recreational activities, such as playing games and watching videos; transactional use, which denotes smartphone use for commercial transactions and online payments; and instrumental use, which represents goal-directed smartphone use for information acquisition, learning, and task completion, such as using navigation applications. When uncertain, we referred to the app's category in the Apple App Store to assign it to one of the four types. Table 2 provides an overview of the basic information and the results of the independent-samples *T*-test for the final sample across both groups. Substantial disparities are evident between the two participant groups regarding problematic smartphone use.

2.2 Measures

2.2.1 Smartphone Addiction Scale (SAS)

The Smartphone Addiction Scale (SAS, Kwon et al. 2013b) adapted a 6-point Likert scale (1 = strongly

Table 2. Demographics and smartphone-related behaviours in two groups.

Variable	PSUs (<i>n</i> = 15)	NPSUs (<i>n</i> = 15)	<i>t</i>	<i>p</i>	Cohen's <i>d</i>
Smartphone addiction (scores)	127.27 ± 24.42	72.13 ± 13.58	7.64	< 0.001	2.79
Smartphone use time (h/day)	9.70 ± 2.70	7.62 ± 2.32	2.09	0.048	0.83
Smartphone use patterns (h/day)					
Interpersonal	5.89 ± 2.11	3.56 ± 1.94	2.92	0.008	1.15
Entertainment	1.92 ± 1.93	1.89 ± 2.1	0.05	0.963	0.01
Transaction	0.27 ± 0.26	0.45 ± 0.72	−0.83	0.415	−0.33
Instrumental use	0.58 ± 0.39	0.56 ± 0.73	0.07	0.945	0.03

Note. Significance level ($p < 0.05$), bold indicates significance.

disagree to 6 = strongly agree), which consists of 33 items, with a higher score indicating higher smartphone addiction levels. It includes six components, daily-life disturbance, positive anticipation, withdrawal, cyberspace-oriented relationship, overuse, and tolerance (Cronbach's $\alpha = 0.957$). Many studies also revealed that the questionnaire demonstrates good reliability and validity within the Chinese population (Ge et al. 2023; Wang, Zhou, and Hu 2024; Xiang, Wang, and Ma 2019).

The Smartphone Addiction Scale-Short Version (SAS-SV, Kwon et al. 2013a) consists of 10 items selected from the full SAS. By extracting these 10 items from participants' responses to the full scale, we obtained their scores on the short version. The SAS-SV was used to screen PSU (Cronbach's $\alpha = 0.913$). Kwon et al. (2013a) determined that a cut-off value of 31 for males and 33 for females was used to define problematic smartphone use. Thus the short version served to classify participants into problematic smartphone use (PSUs) and non-problematic smartphone use (NPSUs), while the full version provided a more comprehensive assessment of problem smartphone use severity across the two groups. As of 2021, the two scales are widely recognised as the most frequently referenced measures for assessing problematic smartphone use (Olson et al. 2022).

2.2.2 State anxiety inventory (S-AI)

The State-Trait Anxiety Inventory (STAI), State version, was adopted to measure state anxiety (Spielberger et al. 1983). The state part of the STAI also consists of 20 statements that assess feeling states (e.g. I feel calm) with Likert-type response options (1 = not at all to 4 = very much so). Over the period of 60 min, the scale was administered three times, each time at intervals of about 30 min, to track the dynamic changes in state anxiety across the experimental session. Each administration is named anxiety 1 (Cronbach's $\alpha = 0.902$), anxiety 2 (Cronbach's $\alpha = 0.886$), and anxiety 3 (Cronbach's $\alpha = 0.906$). The questionnaire has consistently shown strong reliability and validity in Chinese populations (Luo et al. 2018; Shen 1994).

2.2.3 Modified Go/NoGo task

The background cues consisted of smartphone-related and neutral cues. Sixty smartphone app icons were gathered as cues associated with smartphone use, and these were processed using Photoshop CC 2015 to a standardised size of 450×450 pixels. The visual angles were 0.86° (height) for letters and 6.81° for background images. Subsequently, 36 students, uninvolved in the main experiment, assessed the emotional arousal level of all images, rating them from '0 = very unpleasant' to '9 = very pleasant', and gauged their familiarity on a scale of '0 = very unfamiliar' to '9 = very familiar'. The top 45 icons were chosen based on these ratings (mean emotional arousal level = 5.39 ± 0.67 ; mean familiarity level = 6.24 ± 1.25). The smartphone-related cues were expected to possess high familiarity, representing commonly recognised icons, and to have a moderate emotional valence to limit the impact of emotion on inhibitory control ability (Albert, Lopez-Martin, and Carretie 2010). For the neutral background cues, 45 neutral images were sourced from the Chinese Affective Picture System (CAPS, Lu, Hui, and Yu-Xia 2020) and the National Affective Picture System (NAPS, Lang, Bradley, and Cuthbert 1997), with reference to studies by Chen et al. (2016) and Gao et al. (2020). All images were of moderate emotional arousal. Of these, 24 images were from CAPS (mean emotional arousal level = 5.31 ± 0.28), and 21 were from NAPS (mean emotional arousal level = 5.06 ± 0.41).

The target stimuli were capitalised letters 'M' and 'W' (RGB: 255, 255, 0, Arial font). To ensure maximum visibility, each letter appeared against a small black backdrop, which was overlaid onto the background images. Each type of background cue comprised three blocks, making a total of six blocks for the entire task. A Latin-square design was implemented to ensure balanced block presentations amongst participants. Each block had 180 Go trials and 45 NoGo trials in a 4:1 ratio. Every background cue was shown five times: once in the NoGo trial and four times in the Go trials. At the start of each trial, the background cue appeared for 500–600 ms, followed by the target stimulus for 200 ms. This then reverted to an initial grey screen for 1300 ms. Breaks of 2–3 min were provided between

blocks. Participants were guided to press the keyboard letter 'M' as swiftly and accurately as possible when the letter 'M' (Go) was shown on the screen, and to withhold any response when the letter 'W' (NoGo) appeared. Before the primary experiment, participants were given a practice run of 20 trials. A full trial sequence and sample images can be seen in Figure 1.

2.2.4 Electroencephalographic (EEG) recording and analysis

EEG data were captured from 64 scalp sites using tin electrodes affixed to an actiCHamp system (Brain Vision LLC, Morrisville, NC, USA; pass band: 0.01–100 Hz; sampling rate: 1000 Hz). The Fz electrode served as the recording reference. All electrode impedances were maintained below 5 k Ω . During offline analysis, the reference electrode was recalibrated to the mean of the left and right mastoids. Continuous EEG signals underwent a band-pass filter ranging from 0.1 to 40 Hz. The data were then segmented into 1200 ms epochs, extracting 200 ms before and 1000 ms after stimulus onset, and this extracted window was baseline-corrected using the 200 ms interval preceding the stimulus onset. To address electro-oculogram (EOG)

artefacts, an independent component analysis (ICA) algorithm was employed (Jung et al. 2001). Artefact signals with amplitudes exceeding ± 80 μ V were excluded, and such problematic trials accounted for $8.54 \pm 8.9\%$ of all trials.

This study emphasised the N2 and P3 components linked to inhibitory control. N2 component values were ascertained from frontal central electrodes, specifically F3, F1, Fz, F2, and F4, drawing from prior research, with the most negative peak observed approximately 250–350 ms post-stimulus onset. For the P3 component, with its peak positivity observed around 400–500 ms post-stimulus, values were gauged from frontal central electrodes C3, C1, Cz, C2, and C4 (Chen et al. 2020; Donkers and van Boxtel 2004; West and Alain 2000).

2.3 Procedure

Participants were invited to the laboratory, which was maintained at a comfortable temperature and was soundproofed. Upon arrival, they read and signed a consent form and handed over their smartphones to us for safekeeping. We outfitted the participants with

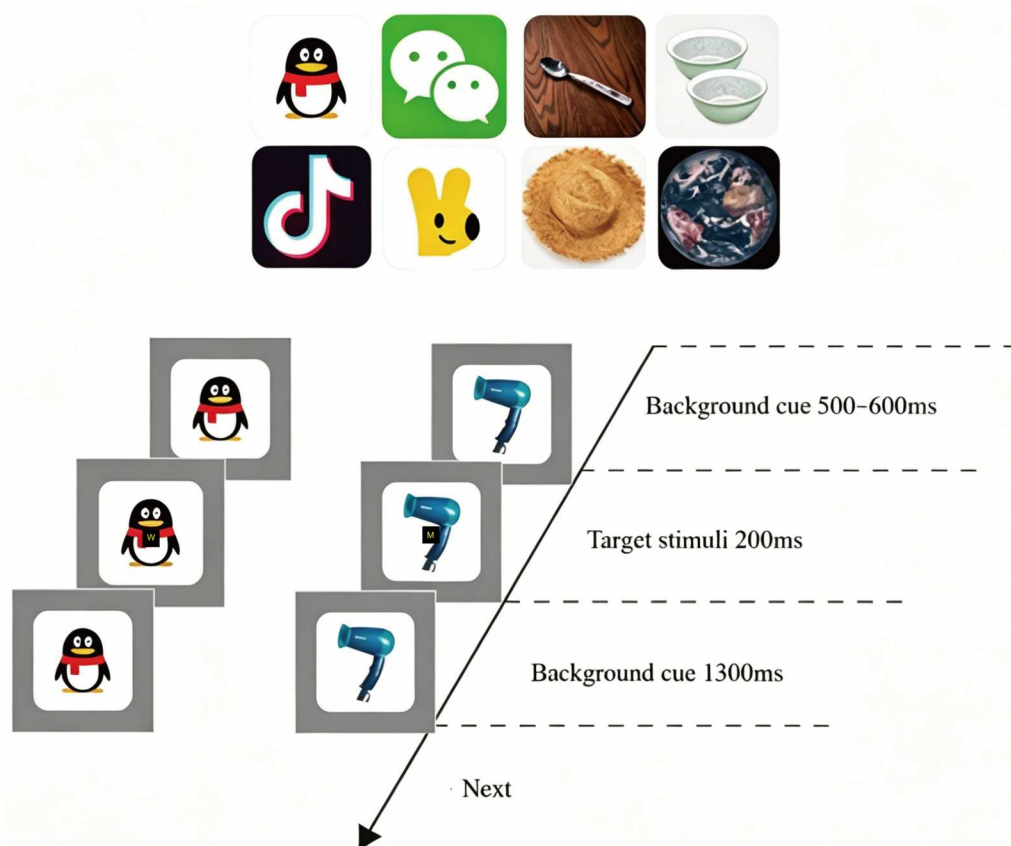


Figure 1. Smartphone (left) and neutral background (right) cues and a completed smartphone cue 'NoGo' (left) and neutral 'Go' (right) trial.

EEG caps for collecting brainwave data. Then, they were asked to sit quietly for 5 min, after which the first state anxiety survey was administered. Following this, participants undertook the modified Go/NoGo task. Upon completion of three blocks (approximately 30 min), participants took a break and the second state anxiety survey was administered. After the Go/NoGo task was completed, participants were handed back their smartphones. The third state anxiety measure was then administered. Following this, the EEG equipment was removed. Participants were then asked to open the screen time statistics panel on their own devices to verify their smartphone usage time, and the experimenter recorded the displayed usage time (four participants' smartphones lacked this feature).

Finally, we expressed our gratitude to the participants and provided a compensation of 50 yuan. The entire procedure is illustrated in Figure 2. The study received approval from the local research ethics committee, and all participants provided their informed consent. All procedures performed in the study involving

human participants were in accordance with the institutional research committee's ethical standards and the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

2.4 Statistical analysis

All analyses were performed using SPSS 23.0. State anxiety scores, obtained at three different time points, were subjected to repeated measures ANOVA. The between-subject factor was the group (PSU, NPSU), while the within-subject factor was the time sequence (Time1, Time2, Time3). In cases where appropriate, Greenhouse-Geisser corrected values were reported.

For the analysis of behavioural data, reaction times of correct responses were examined using a repeated measures 2-way ANOVA with the background cues (smartphone cue, neutral background) as the within-subjects variable, and the group (PSU, NPSU) as the between-subjects variable. Accuracy rates were analysed using a repeated measures 3-way ANOVA with the

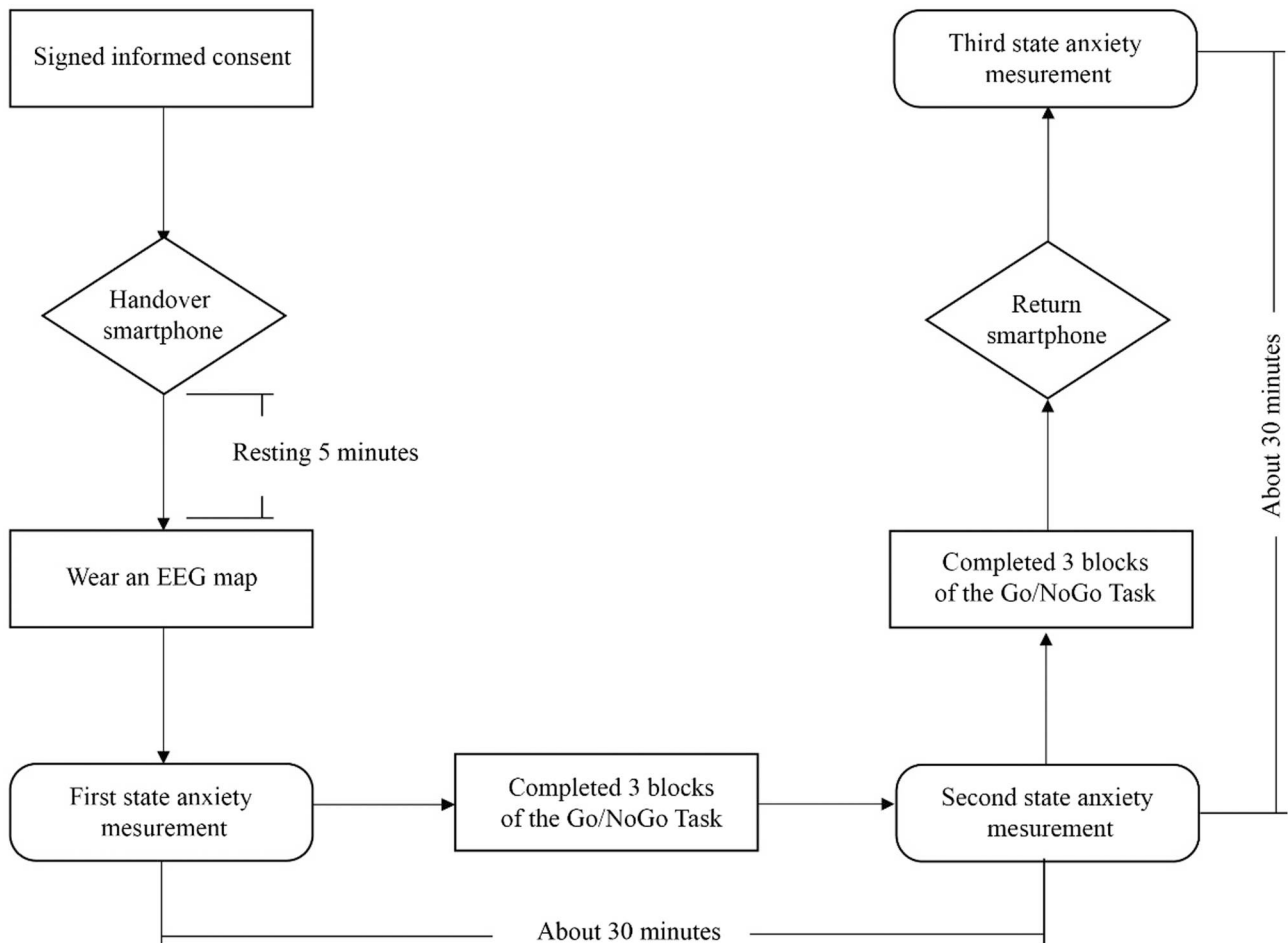


Figure 2. The entire experimental procedure.

factors of condition (Go, NoGo) and background cues (smartphone cue, neutral background) as the within-subjects variables, and the group (PSU, NPSU) as the between-subjects variable.

Regarding the ERP data, the N2 and P3 components were analysed using the same approach as the accuracy rates.

Additionally, independent sample *t*-tests were conducted with the group serving as the independent variable for both behavioural and ERP data, comparing the two groups of subjects.

3. Results

3.1 State anxiety during withdrawal

A two-way ANOVA was performed on state anxiety scores. We found a significant time sequence and group effect. The main effect of time sequence was significant ($F(2, 56) = 3.32, p = 0.043, \eta_p^2 = 0.11$). Post-hoc analyses revealed that the score of state anxiety in time 2 (37.67 ± 1.42) was higher than in time 3 ($34.77 \pm 1.49, p = 0.012$). The main effect of group was significant ($F(1, 28) = 6.46, p = 0.017, \eta_p^2 = 0.19$). Post-hoc analyses revealed that the score of PSU group (39.47 ± 1.80) was higher than the control group (33.00 ± 1.49) ($p = 0.017$). An interaction of time sequence $\times$ group was not significant ($F(2, 56) = 0.38, p = 0.683, \eta_p^2 = 0.01$). The actual trend of anxiety level change is shown in Figure 3.

3.2 Salience-affected inhibitory control ability

3.2.1 Behavioural results

The results of the analysis of variance of behavioural data are shown in Table 3 and Figure 3.

3.2.1.1. Reaction time. The main effect of background cue was significant ($F(1, 28) = 9.38, p = 0.005, \eta_p^2 = 0.25$). Participants responded faster to the smartphone background cue (389.74 ± 9.22 ms) than the neutral (399.37 ± 9.27 ms). No significant differences were found between the groups.

3.2.1.2. Correct rates. The Correct rates were modulated by the main effects of 'condition' ($F(1, 28) = 22.14, p < 0.001, \eta_p^2 = 0.44$) and 'background cues' ($F(1, 28) = 20.42, p < 0.001, \eta_p^2 = 0.42$). Participants demonstrated a higher correct rate in the Go condition ($99.14 \pm 0.55\%$) compared to the NoGo condition ($87.72 \pm 2.44\%$). Moreover, a higher correct rate was observed with the neutral cue ($94.44 \pm 1.17\%$) than with the smartphone background cue ($92.41 \pm 1.44\%$). An

interaction for condition $\times$ background cue was significant ($F(1, 28) = 9.88, p = 0.004, \eta_p^2 = 0.26$). Simple effects analysis indicated that for the NoGo condition, the correct rate in the smartphone background cue ($85.98 \pm 2.69\%$) was lower than the neutral ($89.46 \pm 2.24\%$) ($p < 0.001$), but not for the Go condition. An interaction for condition $\times$ group was significant ($F(1, 28) = 4.25, p = 0.049, \eta_p^2 = 0.13$). Simple effects analysis indicated that for the PSU group, the correct rate of the NoGo condition ($83.18 \pm 3.46\%$) was lower than the Go condition ($99.60 \pm 0.78\%$) ($p < 0.001$), with no differences observed between conditions for the control group. Finally, a triple interaction of condition $\times$ group $\times$ background cues was not significant.

3.2.2 ERP results

Table 4 and Figure 4 show the N2 and P3 amplitudes for various background cues and conditions in two groups. The results of the analysis of variance of ERP data are shown in Table 5.

3.2.2.1. N2. The N2 amplitudes were significantly modulated by the main effects of 'condition' ($F(1, 28) = 17.36, p < 0.001, \eta_p^2 = 0.38$) and 'background cue' ($F(1, 28) = 106.51, p < 0.001, \eta_p^2 = 0.79$). NoGo condition [$(1.71 \pm 0.59) \mu V$] elicited larger N2 amplitudes than Go condition [$(3.83 \pm 0.68) \mu V$]. Smartphone background cue [$(1.25 \pm 0.53) \mu V$] elicited larger N2 amplitudes than the neutral [$(4.28 \pm 0.67) \mu V$]. No significant differences were found between the groups.

3.2.2.2. P3. The P3 amplitudes were significantly modulated by the main effects of 'condition' ($F(1, 28) = 60.37, p < 0.001, \eta_p^2 = 0.68$) and 'background cue' ($F(1, 28) = 27.48, p < 0.001, \eta_p^2 = 0.50$). NoGo condition elicited larger P3 amplitudes [$(10.37 \pm 0.98) \mu V$] than Go condition [$(5.57 \pm 0.63) \mu V$]. Furthermore, the neutral background cue elicited more pronounced P3 amplitudes [$(9.01 \pm 0.83) \mu V$] than the smartphone cue [$(6.93 \pm 0.75) \mu V$]. The 'group' main effect was not significant. A significant triple interaction was observed between condition, group, and background cue ($F(1, 28) = 6.85, p = 0.014, \eta_p^2 = 0.20$). Simple effects analysis indicated that on the smartphone background cue, the NoGo condition evoked smaller P3 amplitudes for the PSU group [$(7.01 \pm 1.41) \mu V$] than the control [$(11.39 \pm 1.41) \mu V$].⁴

4. Discussion

The classification of problematic smartphone use within the spectrum of behavioural addictions is a subject of considerable debate and investigation, engaging diverse

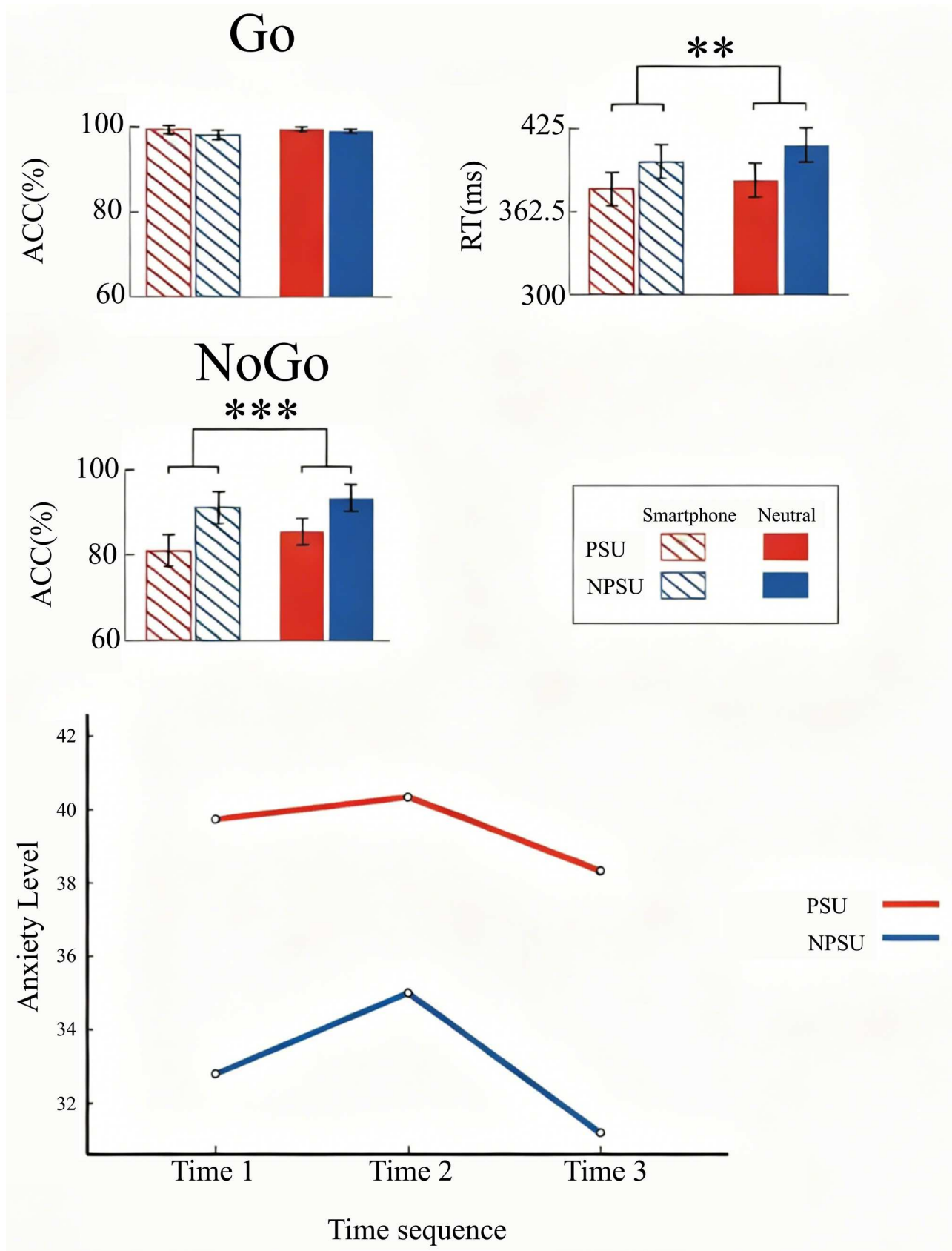


Figure 3. The trend of anxiety level changes and behavioural results, correct rates and reaction time.

Table 3. Analysis of variance results for behavioural data.

Variable	ACC			RT		
	<i>F</i>	<i>p</i>	η_p^2	<i>F</i>	<i>p</i>	η_p^2
Group	2.48	0.127	0.08	1.73	0.199	0.06
Background cue	20.42	< 0.001	0.42	9.38	0.005	0.25
Condition	22.14	< 0.001	0.44	—	—	—
Group × Background cue	0.62	0.439	0.02	0.88	0.356	0.03
Group × Condition	4.25	0.049	0.13	—	—	—
Background cue × Condition	9.88	0.004	0.26	—	—	—
Group × Background cue × Condition	2.71	0.111	0.09	—	—	—

Note. Degrees of freedom (df) (1, 28), significance level ($p < 0.05$), bold indicates significance.

Table 4. N2 and P3 mean amplitude in the two groups.

Background cue	Variables (μV)	PSU ($n = 15$) ($M \pm SD$)	NPSU ($n = 15$) ($M \pm SD$)	<i>t</i>	<i>p</i>
Smartphone	Go-N2 amplitude	2.17 $\pm$ 3.85	2.64 $\pm$ 2.94	−0.37	0.715
	NoGo-N2 amplitude	−0.03 $\pm$ 2.25	0.23 $\pm$ 3.65	−0.23	0.820
	Go-P3 amplitude	4.06 $\pm$ 3.20	5.28 $\pm$ 3.74	−0.96	0.346
	NoGo-P3 amplitude	7.00 $\pm$ 5.19	11.39 $\pm$ 5.68	−2.21	0.036
Neutral	Go-N2 amplitude	4.72 $\pm$ 4.59	5.78 $\pm$ 3.65	−0.70	0.491
	NoGo-N2 amplitude	3.07 $\pm$ 3.30	3.56 $\pm$ 4.04	−0.36	0.718
	Go-P3 amplitude	5.86 $\pm$ 3.41	7.06 $\pm$ 3.71	−0.93	0.363
	NoGo-P3 amplitude	10.88 $\pm$ 6.74	12.21 $\pm$ 4.86	−0.62	0.539

Note. Significance level ($p < 0.05$), bold indicates significance.

scholarly perspectives (Billieux et al. 2015; Elhai et al. 2017; James et al. 2023; Panova and Carbonell 2018; Wang et al. 2016; Wilcockson, Osborne, and Ellis 2019). This study contributes to this ongoing dialogue by empirically examining key aspects associated with behavioural addiction (withdrawal anxiety and inhibitory control impairments) in individuals characterised by their smartphone use. Our investigation aimed to discern whether these manifestations align with the behavioural addiction framework.

In exploring these dimensions, our findings reveal a partial affirmation of this classification: Hypotheses 1a and 2b received support, indicating that individuals with PSU displayed higher anxiety levels during smartphone withdrawal and reduced NoGo-P3 amplitude during background cue trials compared to the control group, respectively. However, Hypothesis 1b, that problematic smartphone users experience more remarkable changes in anxiety levels (compared with NPSU) during separation and reunion with their phones, has not been validated. In addition, Hypothesis 2a was also not supported concerning the early-stage conflict monitoring as reflected by NoGo-N2 amplitudes. There was no significant difference found in NoGo-N2 amplitude between the PSU group and controls during trials with smartphone-related background cues. This introductory discussion will further elaborate on these results, providing a detailed exploration of their implications for understanding problematic smartphone use and whether it has features similar to addictive behaviour.

4.1 Withdrawal anxiety

Withdrawal anxiety, a hallmark symptom of behavioural addiction, manifests as significant changes in anxiety levels following cessation or reduction of addiction-related behaviours (Brown 1993; Goodman 1990; Griffiths 1996, 2005; Panova and Carbonell 2018; Wilcockson, Osborne, and Ellis 2019). Our findings partly resonate with this characterisation, as we observed that problematic smartphone users consistently exhibited higher levels of state anxiety compared to controls throughout the experimental procedure (Hypothesis 1a). The result aligns with the findings of Cheever et al. (2014) and Cutino and Nees (2017), who also reported that participants with more severe problematic smartphone use exhibited higher levels of anxiety during smartphone withdrawal. Caution is nonetheless warranted about the phenomenon; such differences may be attributed to the inherent anxiety trait among problematic smartphone users (Elhai et al. 2017). In other words, even outside of smartphone withdrawal states, such differences may persist between the two groups. Therefore, it is necessary to consider the dynamic changes in anxiety levels between the two groups in conjunction (Hypothesis 1b) to ascertain whether problematic smartphone users experience withdrawal anxiety.

Our findings show that anxiety levels in individuals with problematic smartphone use increase during withdrawal and decrease upon re-access to their devices. A similar pattern was observed in the control group,

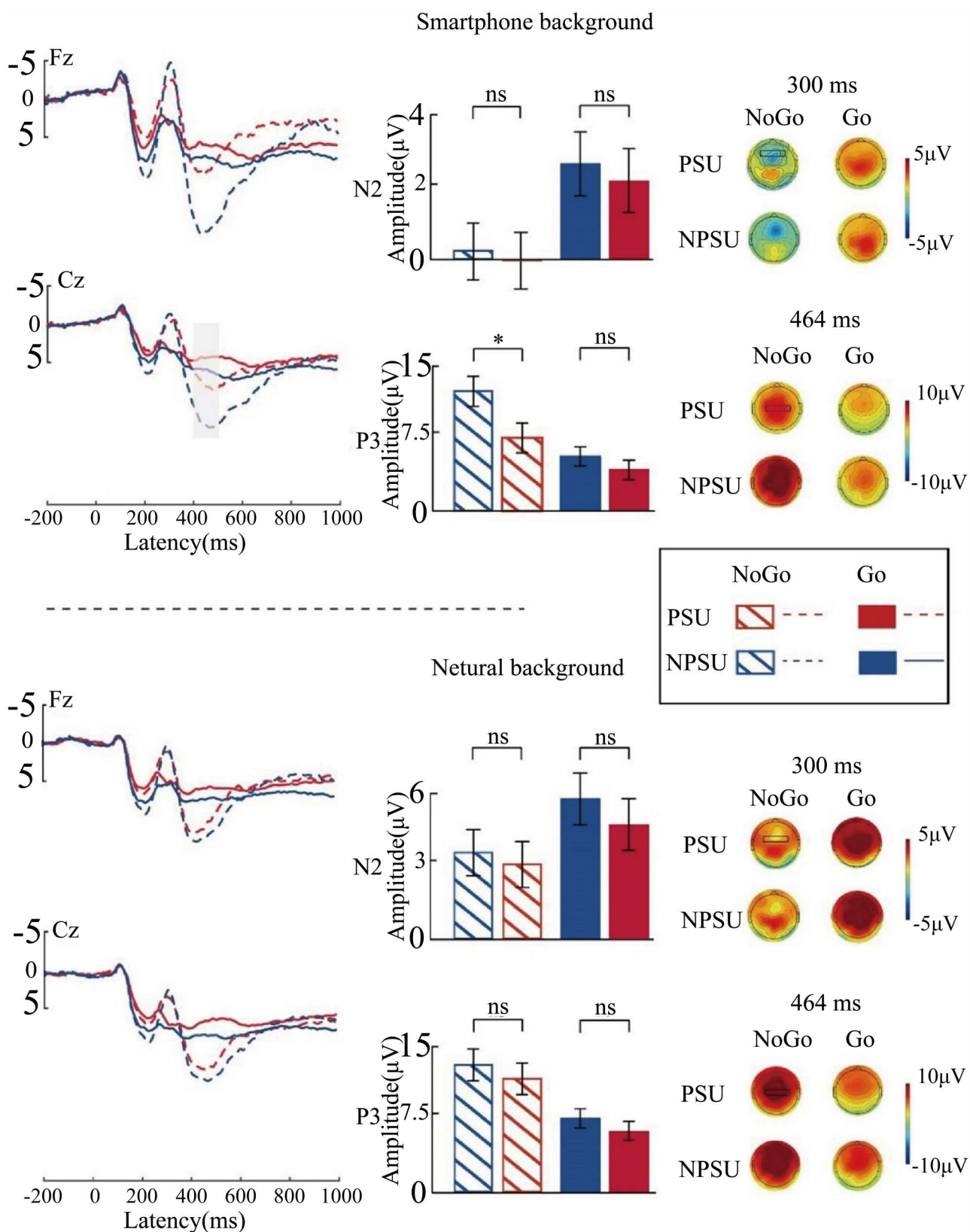


Figure 4. N2 and P3 mean amplitude in the two groups.

Table 5. The results of the analysis of variance of ERP data.

Variable	N2			P3		
	<i>F</i>	<i>p</i>	η_p^2	<i>F</i>	<i>p</i>	η_p^2
Group	0.24	0.631	0.01	1.77	0.194	0.06
Background cue	106.51	< 0.001	0.79	27.48	< 0.001	0.50
Condition	17.36	< 0.001	0.38	60.36	< 0.001	0.68
Group × Background cue	0.50	0.484	0.02	3.75	0.063	0.12
Group × Condition	0.14	0.707	0.01	1.77	0.194	0.06
Background cue × Condition	1.56	0.222	0.05	0.92	0.345	0.03
Group × Background cue × Condition	0.38	0.540	0.01	6.85	0.014	0.20

Note. Degrees of freedom (df) (1, 28), significance level ($p < 0.05$), bold indicates significance.

with even more pronounced changes (see Figure 3). Thus, Hypothesis 1b, which predicted that the control group would show less variation – was not supported. This result can be explained from two perspectives. First, individuals with problematic smartphone use tend to exhibit chronically elevated anxiety levels, which may leave little room for further fluctuation. Second, smartphones play a crucial role in daily life for all individuals, and their absence can induce anxiety regardless of problematic use status – an explanation consistent with the theory of self-extension (Belk 1988, 2016). According to this theory, personal objects that individuals deeply integrate into their self-concept become extensions of their identity. Being deprived of such devices may feel as disabling as losing a limb, restricting one's capacity to interact with the world and, in turn, elevating anxiety levels. Given the ubiquity of smartphone use in contemporary society, individuals – irrespective of whether they exhibit problematic use – tend to incorporate smartphones as extensions of the self. Consequently, smartphone withdrawal is likely to increase anxiety across populations. These findings support the extended self-theory in understanding the psychological effects of smartphone use.

4.2 Salience-induced inhibitory control impairment

We also investigated whether the salience of smartphone cues influences inhibitory control in individuals with problematic smartphone use. The analysis of behavioural data demonstrated marked variations in accuracy between Go and NoGo trials among the PSU group, whereas such differences were not observed in the control group. This finding suggests that problematic smartphone users face difficulties in inhibiting their dominant responses, which is consistent with previous research (Zhou and Bai 2023). Furthermore, impairments in inhibitory control are widely recognised as prominent behavioural manifestations of impulsivity, and it is well-established that impulsivity accompanies problematic smartphone use (Gansler et al. 2011;

Garavan 2011; Gopal et al. 2013; Tan et al. 2024). Impulsivity is a well-established predictor and commonly observed behavioural characteristic in substance addiction (Dalley and Ersche 2019; Long et al. 2018; Zeng et al. 2022).

Notably, our analysis did not reveal significant differences in cue salience-induced inhibitory control impairment at the behavioural level between the two groups, which could be attributed to a speed-accuracy trade-off. Participants might prioritise either response speed or accuracy depending on the task demand or their strategy, potentially obscuring differences in inhibitory control.

4.2.1 N2 component and conflict monitoring

Despite the anticipation of observable differences in the N2 component (Hypothesis 2a), indicative of conflict monitoring and effortful processing through the anterior cingulate cortex (ACC) activities (Chiu, Holmes, and Pizzagalli 2008; Zhang and Lu 2012), our analysis revealed no significant differences between groups. This outcome aligns with previous findings (Chen et al. 2016; Gao et al. 2020; Petit et al. 2012). Delving into the roles of ERN and N2 in conflict monitoring, we consider two explanations for this discrepancy. First, the ACC's involvement might predominantly manifest in the ERN component, immediate to erroneous responses (Falkenstein et al. 1990; Gehring et al. 1993), rather than in N2, which is more related to processing irrelevant information. This suggests a potential phase of conflict monitoring discrepancy that could be more pronounced in the ERN stage (Yeung and Cohen 2006). Second, the habitual engagement with irrelevant stimuli, characteristic of problematic smartphone users, could diminish the expected N2 amplitude differences, indicating a predisposition toward habitual processing (Gao et al. 2020).

4.2.2 P3 component: assessing inhibitory control

Contrastingly, the analysis of the P3 component significantly supported Hypothesis 2b, indicating impaired later stages of inhibitory control among problematic smartphone users when exposed to smartphone-related

cues. This finding aligns with previous studies (Gao et al. 2020) and the impaired response iRISA model's predictions (Goldstein and Volkow 2002, 2011) and underscores the challenges in later stages of inhibitory control that problematic smartphone users face, particularly in the presence of salient cues. This is further supported by neuroimaging studies indicating structural and functional alterations in the orbitofrontal cortex (OFC) of problematic smartphone users (Lee et al. 2019), which is implicated in decision-making and impulse control (Torregrossa, Quinn, and Taylor 2008).

Findings from Gao et al. (2020) and the present study diverge from those reported by Chen et al. (2016). We attribute this inconsistency to profound changes in smartphone technology and usage ecosystems that have occurred since 2016, when Chen et al. completed their data collection. Specifically, smartphone development has gradually shifted from passive response to active intelligent engagement, mainly in four aspects: First, smartphones have evolved into always-available, multi-purpose hubs integrating work, entertainment, and social communication (Imgraben, Engelbrecht, and Choo 2014; Oh, Syn, and Jang 2020). Second, notification systems have become far more intrusive and attention-grabbing (Firth et al. 2019; Fournier et al. 2026). Third, app designs widely adopt features such as infinite scrolling and gamification to foster habitual use and enhance user retention (Hasan, Jha, and Liu 2018; Tam and Inzlicht 2024; Tian, Bi, and Chen 2023). Fourth, most smartphone apps now incorporate algorithm-driven mechanisms designed to maximise user dwell time and reinforce sustained usage (Hasan, Jha, and Liu 2018). Notably, this technological trajectory continues to intensify. Technology is not neutral: it continuously captures user attention, encourages high-frequency use, and seeks to convert attention into economic value (Tian, Bi, and Chen 2023). Together, these changes have greatly increased the salience of smartphone-related cues, imposing greater demands on the salience-based inhibitory control of individuals with PSU. In contrast, when Chen et al. conducted their study around 2016, such technologies and features remained underdeveloped. Consequently, smartphone-related cues lacked sufficient salience to tax inhibitory control or reliably reveal group differences. In summary, our findings not only replicate the iRISA model but also extend it by demonstrating that technological evolution may amplify salience-induced inhibitory impairments.

4.2.3 Integrative discussion

These findings collectively provide a nuanced comprehension of the cognitive neural mechanisms associated with problematic smartphone usage. The absence of

significant differences in N2 amplitudes suggests a complex dynamic in conflict monitoring among individuals with problematic smartphone usage, a topic ripe for further exploration.

Meanwhile, the significant findings regarding the P3 component provide empirical support for the impaired response inhibition and iRISA model's applicability to problematic smartphone usage, illustrating how addiction-cue salience undermines effective inhibitory control. This suggests shared neurobiological underpinnings across various forms of addiction and provides preliminary evidence for the classification of problematic smartphone usage within the behavioural addiction framework. These insights into the neurocognitive and neurobiological aspects of problematic smartphone usage underscore the importance of considering both cognitive control deficits and the broader neural mechanisms at play.

5. Conclusion

Overall, this study explored the classification of problematic smartphone use within the spectrum of behavioural addictions by examining withdrawal anxiety and salience-induced inhibitory control impairment in individuals with problematic smartphone use. Our findings partially affirm the classification of problematic smartphone use as a behavioural addiction. Specifically, individuals with problematic smartphone use displayed higher levels of anxiety during smartphone withdrawal and impaired later stages of inhibitory control when exposed to smartphone-related cues. However, not all hypotheses were supported. The PSU group did not experience a sharp rise in anxiety upon separation from their phones, nor did it follow with a rapid decline upon reunion. In addition, there was intact early stage of inhibitory control when the PSU group was exposed to smartphone-related cues, which is contradictory to our hypothesis.

These findings validate the application of the iRISA model and the Extended Self Theory to problematic smartphone users. As we continue to navigate the complexities of digital technology's impact on human behaviour and brain function, our study offers pivotal insights into the mechanisms of problematic smartphone usage, advocating for targeted interventions that address both the psychological and neurobiological facets of this pervasive issue.

6. Implications, limitations and future research

This study focuses on withdrawal anxiety and salience-induced inhibitory control impairment, with the aim of

contributing modestly to the ongoing debate over whether PSU qualifies as a behavioural addiction. Beyond this core theoretical question, the present work also provides several meaningful empirical and practical insights. First, we employed a more rigorous participant screening protocol by integrating the established criteria from both Gao et al. (2020) and Chen et al. (2016). This approach allows our findings to characterise the psychological profile of PSU individuals with greater reliability. Second, prior ERP studies have yielded inconsistent conclusions regarding the nature of inhibitory control deficits in PSU: Chen et al. supported a general inhibitory deficit, whereas Gao et al. argued for a cue-specific impairment. Our findings lend greater support to the cue-specific account, which may help reconcile these conflicting results. Third, the present study was conducted nearly a decade after Chen et al. (2016) and several years following Gao et al. (2020). Over this period, smartphone technology and usage ecology have changed considerably. Re-examining inhibitory control under this updated context, we found that individuals with PSU still display salience-induced inhibitory impairment. This suggests that, alongside technological evolution, PSU may increasingly exhibit core features of behavioural addiction. Fourth, our results confirm that the temporary removal of smartphones elicited anxiety in all participants, providing empirical evidence for the pervasive embeddedness of digital technology in everyday life. On the practical side, this study offers preliminary empirical support for digital detox interventions: reducing exposure to smartphone-related cues, for instance, may help attenuate problematic smartphone use. For developers of apps designed to limit excessive smartphone use, our findings further suggest that forced lock-outs or abrupt disconnection may exacerbate anxiety among general users. By contrast, designs that enable gradual disengagement, retain access to essential information, and employ low-arousal notifications may better strike a balance between curbing overuse and maintaining psychological well-being.

Despite these contributions, our investigation encounters several limitations that warrant careful consideration. Our analysis was constrained to two specific criteria of behavioural addiction, offering a focused yet partial examination of problematic smartphone usage. The study, focused on a college population with a relatively small sample size, limited the generalisability and robustness of our findings, therefore it should be regarded as a preliminary exploration (pilot study) for more extensive research with larger samples. Additionally, we did not record the exact time each participant took for EEG cap fitting, which may have introduced

minor variability in the duration of smartphone separation prior to the first anxiety assessment. Furthermore, relying exclusively on self-report questionnaires without electrophysiological measures of withdrawal anxiety may fail to adequately dissociate withdrawal effects from trait anxiety.

The preliminary nature of this research underscores the necessity for further exploration. Future studies should aim to include a broader array of criteria for behavioural addiction, providing a more holistic view of problematic smartphone usage. Employing larger, more diverse sample sizes is crucial for enhancing the generalisability of findings. Furthermore, extending the duration of smartphone abstinence could yield deeper insights into the temporal dynamics of withdrawal effects. In addition, longitudinal research designs could further elucidate the progression and long-term impacts of problematic smartphone use on neurocognitive functions. Moreover, using electrophysiological measures to explore situation-based state anxiety among problematic smartphone users is a research path worth considering. Additionally, a more in-depth investigation should be conducted into specific patterns of PSU, such as excessive checking and prolonged usage, as these distinct usage patterns may have varied impacts on individual inhibitory control functions at different stages.

An interdisciplinary approach, integrating insights from psychology, neuroscience and digital technology design, could offer comprehensive solutions to the challenges posed by problematic smartphone usage. Such collaborations might lead to the development of more effective prevention and treatment strategies, grounded in a deep understanding of the addiction's multifaceted nature. Moreover, situating problematic smartphone usage within broader societal and cultural contexts could enhance our comprehension of its prevalence and impact, considering how digital connectivity and societal norms around smartphone use contribute to the problematic usage phenomenon.

Compliance with ethical standards

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The study was approved by the University of the first author.

Disclosure statementNo potential conflict of interest was reported by the author(s).

Author contributions

CRedit: **Jiamin Ge**: Conceptualization, Data curation, Methodology, Writing – original draft; **Anbang Zhang**: Supervision, Writing – review & editing.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon request.

Informed consent

Informed consent was obtained from all individual adult participants included in the study.

Author contribution statement

Jiamin Ge: Conceptualisation, data curation, methodology, writing – original draft. Anbang Zhang: writing – review and editing, supervision.

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